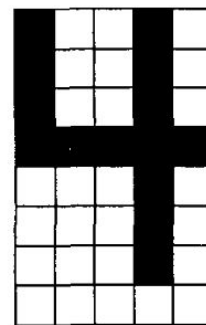
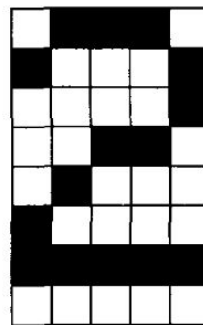
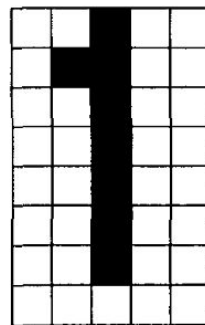

Intro Deep Learning Session 2

by adsoft

Red HopField

In our C++ program we consider synchronous operations. We consider the three patterns given by the numbers 1, 2 and 4 in the figure. A black box indicates +1 and a white box -1. Thus in our example we have $N = 40 = 8 \cdot 5$ neurons, i.e., we have 8 rows and 5 columns in our grid.



Hopfield found experimentally that the number of binary patterns p that can be stored and recalled in a net with reasonable accuracy, is given approximately by

$$p \approx 0.15N$$

where N is the number of neurons in the net. Another estimate is

$$p \approx \frac{N}{2 \log_2 N}.$$

For $N = 40$ Hopfield's estimation is $p = 6$.



Challenge

Programar en Python la red Hopfield para reconocer **figuras** en una matriz de $M \times N$ usando arrays, ciclos for, while y sentencias if, else.

**NOTA: no usar modulos numpy.
Tensorflow, pytorch, etc.**



crear un repo en github.com

subir proyecto a repo

ejemplo:

<https://github.com/adsoftsito/hopfield>